

FoldShield⁺⁺ Structural Analysis Report

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Input Structures

Reference PDB	1UBQ.pdb
Query PDB	1UBI.pdb
Mode	ca
Ref chain	A
Query chain	A

Classical Coordinate Metrics

These metrics saturate near 1.0 even when function diverges. Use FoldShield++ signals below for structural perturbation insight.

Metric	Value	Note
TM-score	1.000	>=0.50 same fold · >=0.90 nearly identical
RMSD (A)	0.100	Lower is better · >3.0 A = significant deviation

FoldShield++ Signal Scores — Phase 1 + Phase 2

All active signals with scores, fusion weights, and interpretation. Phase 2 signals are highlighted in blue.

Braid similarity	0.994	0.20	braid sequences nearly identical
Motif similarity (UL-DSL)	0.912	0.37	nearly identical motif architecture
SSEF entropy (Phase 2) [P2]	0.958	0.20	symbolic grammar nearly identical — buried-core mutation or identical
Persistent homology (Phase 3) [P2]	0.982	0.23	persistent homology features nearly identical

Fusion Layer v2 — Combined Score

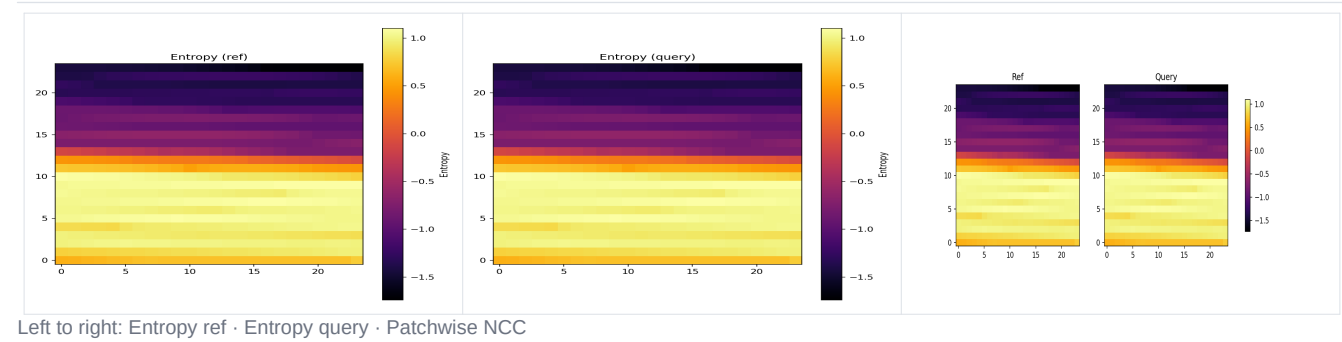
0.9540	phase2	nearly identical protein structures

Phase 2 — Perturbation Classification

SSEF score	Local topology	Gap score	Perturbation class
0.958	0.888	0.070	Point mutation (buried core) or identical

Gap = SSEF minus local topology. <0.15 buried-core mutation | 0.15-0.36 fold-switch | >0.36 conformational transition

Entropy Maps & Signal Profiles



Signal Score Bars

Visual 0-1 scale for all active signals. Green ≥ 0.66 / Amber 0.33-0.66 / Red < 0.33 .



Symbolic Score Detail

Symbolic Similarity Score	0.994	overall nearly identical
Behavioural Correlation Similarity	1.000	entropy maps almost identical (strong dynamic match)
Structural Topology Similarity	1.000	identical set of generators
Structural Topology Ordering Similarity	0.975	patterns & local motifs nearly identical

Signal Analysis — Figures

